

Mtech Software Engineering

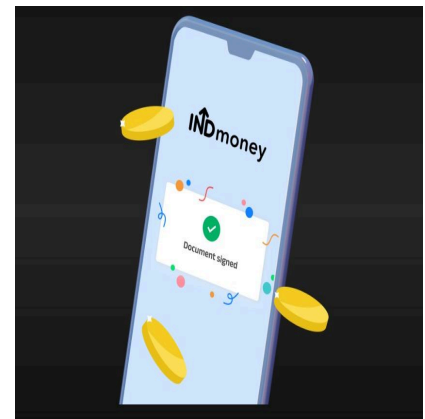
Software Quality Assurance and Testing (S2-25_SEZG501)



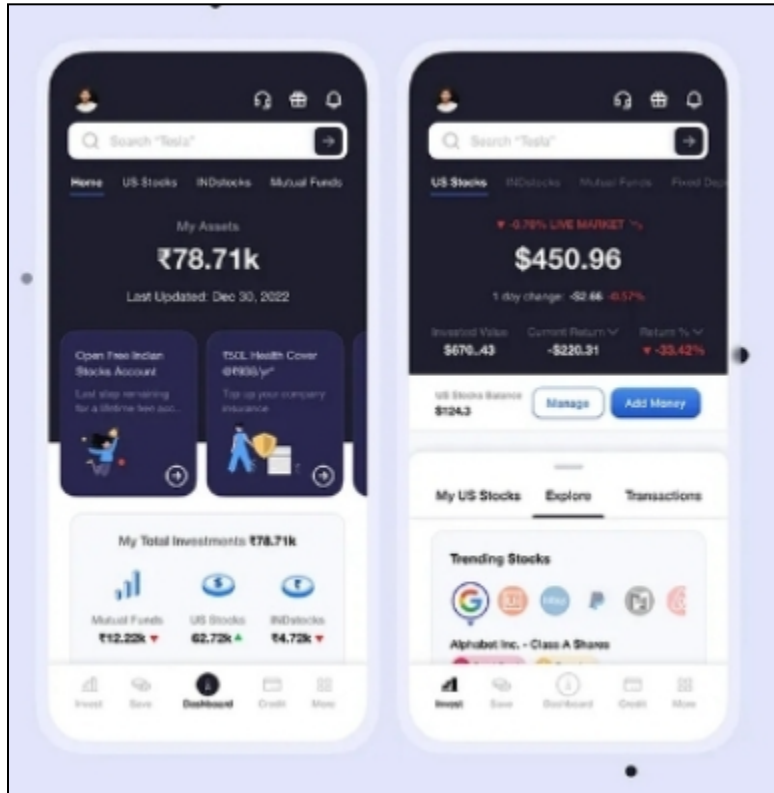
The objective is to strategically apply SQA principles to ensure a high-criticality finance app meets all stated and implied needs through process engineering.

Group Members:

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SQA Responsibilities for Mobile App :



*Name of the Application: **INDMoney***

*Primary Category: **Investment / Wealth Management App***

*Tagline: **More than an Investing App - Your Money Operating System***

Track all your investments, manage family accounts, and see your complete net worth in one place

Introduction:

In this assignment, we analyze the Software Quality Assurance (SQA) responsibilities for a finance-based mobile application, with a specific focus on INDmoney. The objective is to understand how SQA ensures the delivery of a reliable, secure, and high-performance application in the financial domain, where accuracy and trust are critical.

INDmoney is a premier "WealthTech" Super App in the Indian financial landscape, designed to act as an all-in-one financial aggregator and investment platform. Unlike traditional banking apps, INDmoney architecture relies on complex integrations with multiple external entities, including NSDL/CDSL for share tracking, EPFO for retirement funds, various Indian banks for balance fetching, and international brokers for US Stock trading. From an SQA standpoint, this product is classified as a High-Criticality System, where even minor software defects can lead to significant financial discrepancies, regulatory non-compliance, and loss of consumer trust.

The application's core value proposition lies in its ability to provide a "Single Pane of Glass" view of a user's entire net worth. This involves high-stakes data orchestration, where INDmoney must parse diverse data formats ranging from real-time API feeds to legacy email statements while maintaining absolute mathematical precision. Consequently, Software Quality Assurance for this product shifts from simple bug-finding to a proactive Process Engineering approach, ensuring that every integration point is robust, every currency conversion is accurate, and every user data point is protected by bank-grade security protocols.

In this assignment, we examine the SQA lifecycle of INDmoney by analyzing its technical features and quality requirements. The focus is on moving beyond functional testing to address the "Implied Needs" of the user, such as system availability during global market hours and data integrity across cross-border transactions. By implementing rigorous SQA functions and specific process activities, we aim to mitigate the Business Risks inherent in financial technology and establish a reliable "Go/No-Go" framework for software release.

Key Feature Highlights (Customer Point of View):

- Unified Financial Dashboard: View all investments, assets, and liabilities in one place
- Real-time Portfolio Tracking: Instant updates on stock, mutual fund, and US market investments
- Smart Insights & Analytics: AI-driven suggestions for better financial decisions

Generic SQA functions:(1st deliverable)

Generic Software Quality Assurance (SQA) functions play a crucial role in ensuring that a finance application like INDmoney delivers a reliable, secure, and high-quality user experience throughout its lifecycle. These functions include activities such as quality planning, requirements validation, process standardization, testing, defect management, and continuous monitoring

SQA Function	Objective	Key Activities / Application in INDmoney
1. Quality Planning	Define quality goals, standards, and metrics for the project	Establish quality benchmarks for financial accuracy, security, and performance of portfolio tracking and transactions
2. Requirements Quality Assurance	Ensure requirements are clear, complete, and testable	Validate customer requirements like real-time updates, multi-asset tracking, and secure login features
3. Process Definition & Improvement	Standardize development and QA processes	Adopt Agile/Scrum practices, CI/CD pipelines for continuous improvement in feature delivery
4. Code Quality Assurance	Maintain high-quality, maintainable, and secure code	Conduct code reviews, enforce coding standards, and check for vulnerabilities in financial transactions
5. Testing Strategy & Management	Ensure proper test planning and execution	Define unit, integration, system, and UAT testing for features like stock trading and expense tracking

6. Defect Management	Identify, track, and resolve defects efficiently	Use bug tracking tools to monitor issues in real-time data sync, payment failures, or UI glitches
7. Security Assurance	Protect user data and prevent unauthorized access	Validate encryption, secure APIs, and authentication mechanisms (OTP, biometrics)
8. Performance & Reliability Assurance	Ensure app performs well under various loads	Test app responsiveness during peak trading hours and ensure uptime for real-time updates
9. Configuration & Change Management	Manage changes without affecting stability	Control versioning of app releases, APIs, and database updates for seamless feature upgrades
10. Compliance & Audit Assurance	Ensure adherence to financial and regulatory standards	Validate compliance with financial regulations, data privacy laws, and audit trails for transactions

Specific Features of INDmoney (2nd deliverable)

A key specific feature of INDmoney is its ability to act as an **all-in-one financial super app**, combining investment, tracking, and financial planning into a single platform. It enables users to manage stocks, mutual funds, US investments, expenses, and net worth seamlessly through a unified interface. From a customer perspective, this integration provides convenience, better financial visibility, and smarter decision-making without switching between multiple apps

1. Unified Financial Dashboard (Net Worth Tracking):

INDmoney enables users to connect multiple financial accounts—bank accounts, stocks, mutual funds, credit cards, loans—and view everything in a **single consolidated dashboard**. This eliminates the need to switch between multiple apps and provides a real-time snapshot of overall net worth.

Customer Value:

Users gain **complete financial visibility in one place**, helping them make better decisions and track wealth growth effortlessly.

SQA Focus:

- Data accuracy across multiple integrations
- Real-time sync without inconsistency

2. Investment in Indian & US Stocks:

The platform allows seamless investment in both **Indian and international (US) stock markets** with easy onboarding and minimal barriers. Users can buy fractional shares and diversify globally.

Customer Value:

- Access to global markets from a single app
- Portfolio diversification and growth opportunities

SQA Focus:

- Accurate currency conversion and pricing
- Real-time market data and transaction reliability

3. Mutual Funds & SIP Automation:

INDmoney offers **direct mutual fund investments with zero commission** and supports automated SIPs (Systematic Investment Plans) for disciplined investing.

Customer Value:

- Hassle-free automated investments
- Cost-effective investing with better returns

SQA Focus:

- Accurate SIP scheduling and execution
- Reliable transaction processing

4. Goal-Based Financial Planning:

Users can define financial goals like **retirement, buying a house, or education**, and link investments to track progress toward these goals.

Customer Value:

- Structured financial planning
- Clear roadmap for achieving life goals

SQA Focus:

- Accuracy of projections and calculations
- Consistency in progress tracking

5. Smart Insights & Portfolio Analytics (INDInsights):

The app provides **AI-driven insights**, portfolio health analysis, and recommendations to optimize investments.

Customer Value:

- Data-driven decision-making
- Personalized investment suggestions

SQA Focus:

- Algorithm correctness and reliability
- Relevance and accuracy of insights

6. Family Account Aggregation:

Users can **track and manage financial data of family members** under a single account view, making it easier to manage household finances.

Customer Value:

- Centralized family wealth management
- Better financial planning for dependents

SQA Focus:

- Strong access control and privacy mechanisms
- Secure multi-user data handling

7. Advanced Trading Tools & Alerts:

INDmoney provides features like **real-time stock trading, watchlists, price alerts, and market insights**, helping users react quickly to market changes.

Customer Value:

- Faster and informed decision-making
- Better control over investments

SQA Focus:

- Low latency and high performance
- Accuracy of alerts and execution timing

8. Instant Loans & Credit (InstaCash):

The app offers **quick loans and credit options** based on user eligibility and existing assets, providing liquidity without selling investments.

Customer Value:

- Immediate access to funds
- Financial flexibility during emergencies

SQA Focus:

- Secure and compliant financial processing
- Accuracy in eligibility and loan calculations

9. Secure & Paperless Onboarding:

INDmoney supports **fully digital onboarding with KYC verification**, ensuring users can start investing quickly and securely.

Customer Value:

- Fast and convenient account setup
- Trust in data security and privacy

SQA Focus:

- Strong authentication and encryption
- Compliance with regulatory standards

While **brainstorming the specific features of INDmoney**, we can think from a user's daily financial journey and raise practical questions: How easily can a user view all their investments in one place? Is the unified dashboard always showing accurate and real-time data? Can users seamlessly invest in Indian and US stocks without complexity? Are SIPs automated correctly every time? How reliable are alerts and insights for making decisions? Is the app simple enough for beginners but powerful for advanced users?

Considering these scenarios helps identify important features like **unified net worth tracking, global investing, smart insights, goal-based planning, and secure transactions**. This kind of discussion ensures that features are not just available, but also meaningful, reliable, and valuable from a real customer perspective.

Final Insight

These features collectively make INDmoney a **comprehensive financial super app**. From an SQA perspective, the challenge lies in ensuring **accuracy, security, performance, and seamless integration**, as even minor defects can directly impact user trust and financial outcomes.

Quality Requirements for INDmoney (Enhanced & Structured)

(3rd deliverable)

1. Data Accuracy & Integrity:

The system must ensure **highly precise and consistent financial data aggregation, calculations, and reporting** across all integrated sources such as bank accounts, stock markets, mutual funds, and credit systems. Since users rely on this data for critical financial decisions, even minor discrepancies can lead to loss of trust.

- **Expectation:** No mismatch in portfolio valuation, balances, or transaction records across systems
- **Metric:** $\geq 99.99\%$ data accuracy and consistency
- **SQA Validation:** End-to-end data reconciliation, API response validation, database integrity checks, cross-platform verification

2. Security & Data Privacy:

The application must provide **robust security mechanisms to protect sensitive financial and personal data** from breaches, fraud, and unauthorized access. This includes encryption, secure authentication, and compliance with privacy standards.

- **Expectation:** Complete protection of user identity, financial data, and transaction details
- **Metric:** Zero critical/high vulnerabilities; secure authentication success rate $\geq 99.9\%$
- **SQA Validation:** Penetration testing, vulnerability scanning, encryption validation, multi-factor authentication testing

3. Performance & Scalability:

The system should deliver **fast, responsive, and scalable performance**, even during peak usage times such as stock market hours or high transaction volumes.

- **Expectation:** Smooth user experience with minimal latency under all load conditions
- **Metric:** Response time < 2 seconds for key operations; supports high concurrent users
- **SQA Validation:** Load testing, stress testing, scalability testing, performance benchmarking

4. Reliability & High Availability:

The application must ensure **continuous availability and dependable operation**, minimizing downtime and ensuring consistent access to financial services.

- **Expectation:** Users can access the app anytime without disruptions
- **Metric:** $\geq 99.9\%$ system uptime; minimal failure rate
- **SQA Validation:** Failover testing, disaster recovery testing, uptime monitoring, redundancy validation

5. Real-Time Data Synchronization:

The system must provide **real-time or near real-time synchronization of financial data**, ensuring that users always see the latest information regarding their investments and transactions.

- **Expectation:** Instant updates for transactions, stock prices, and portfolio changes
- **Metric:** Data latency $\leq$ few seconds
- **SQA Validation:** API latency testing, real-time data flow validation, synchronization testing across services

6. Usability & User Experience (UX):

The application should offer a **simple, intuitive, and user-friendly interface** that caters to both novice and experienced users, ensuring ease of navigation and interaction.

- **Expectation:** Minimal learning curve with clear, accessible features
- **Metric:** User satisfaction score $\geq 4/5$; reduced user errors
- **SQA Validation:** Usability testing, accessibility checks, user journey validation, feedback analysis

7. Compliance & Regulatory Adherence:

The system must strictly adhere to **financial regulations, KYC norms, and data protection laws**, ensuring legal compliance in all operations.

- **Expectation:** Full compliance with regulatory frameworks and audit requirements
- **Metric:** 100% compliance; zero regulatory violations
- **SQA Validation:** Compliance audits, documentation verification, regulatory testing

8. Maintainability & Continuous Improvement:

The application should be designed for **easy maintenance, quick updates, and continuous enhancements** without impacting existing functionalities.

- **Expectation:** Faster bug fixes and seamless feature updates
- **Metric:** Reduced defect resolution time; high code maintainability index
- **SQA Validation:** Code reviews, CI/CD pipeline validation, regression

Final Insight

These quality requirements ensure that INDmoney delivers a **trusted, secure, high-performing, and scalable financial ecosystem**. From an SQA perspective, maintaining these standards is essential because the application directly handles **critical financial data, real-time transactions, and user trust**, making quality assurance a core success factor.

Process Activities for Ensuring Quality INDmoney (4th deliverable)

To successfully achieve the defined **quality requirements**, a well-structured set of **process activities** must be implemented across development, testing, and management layers. In a finance application like INDmoney where **data accuracy, security, and real-time performance are critical** these activities are not optional but essential. As an SQA, the responsibility is to **define, monitor, audit, and continuously improve** these processes using measurable quality metrics.

1. Requirements Engineering & Quality Validation:

This process ensures that all requirements are **clear, unambiguous, testable, and aligned with customer expectations and financial regulations** before development begins.

- **Development Team:** Analyze feasibility, identify dependencies (APIs, banking integrations), and refine technical requirements

- **Test Team:** Derive test cases, define acceptance criteria, and identify edge cases early
- **Managers:** Approve finalized scope, ensure requirement traceability and prioritization
- **SQA Role:** Conduct formal requirement reviews, eliminate ambiguities, maintain RTM (Requirement Traceability Matrix), and ensure coverage of all financial scenarios (transactions, failures, edge cases)

2. Secure Coding & Development Standards:

Focuses on building **robust, secure, and maintainable code**, especially critical for handling financial transactions and sensitive user data.

- **Development Team:** Follow secure coding practices (input validation, encryption, API security), modular design, and version control
- **Test Team:** Perform static code analysis and validate security at early stages
- **Managers:** Enforce coding standards, peer reviews, and technical governance
- **SQA Role:** Audit code quality, track adherence to standards.

3. Continuous Integration & Automated Delivery (CI/CD):

Ensures **frequent integration of code changes with automated validation**, reducing integration risks and improving release confidence.

- **Development Team:** Commit code frequently, resolve integration conflicts quickly
- **Test Team:** Integrate automated test suites (unit, API, regression) into pipelines
- **Managers:** Ensure CI/CD pipeline stability and infrastructure readiness
- **SQA Role:** Monitor build success rates, enforce automated quality gates, track test coverage, and ensure no defective code progresses to later stages

4. End-to-End Testing & Quality Validation Lifecycle:

Ensures that all features of INDmoney are validated across **functional, integration, system, regression, performance, and security levels**.

- **Development Team:** Conduct unit testing and support defect fixing
- **Test Team:** Execute comprehensive testing including real-time data validation, transaction flows, and edge cases
- **Managers:** Allocate resources and ensure timely completion of testing cycles
- **SQA Role:** Monitor test effectiveness, track defect density, ensure high coverage, and validate that financial calculations and workflows are error-free

5. Defect Management & Root Cause Analysis (RCA):

A structured approach to ensure **early detection, proper tracking, and permanent resolution of defects**.

- **Development Team:** Fix defects and perform root cause analysis to prevent recurrence
- **Test Team:** Log defects with complete details (steps, severity, impact)
- **Managers:** Prioritize defects based on business impact and ensure closure timelines
- **SQA Role:** Analyze defect trends, identify leakage (escaped defects), validate RCA effectiveness, and ensure continuous improvement in quality

6. Performance Engineering & Scalability Assurance:

Ensures the system can handle **high user loads, real-time stock updates, and peak transaction volumes** without degradation.

- **Development Team:** Optimize backend services, APIs, and database queries
- **Test Team:** Perform load, stress, and endurance testing simulating real-world use
- **Managers:** Ensure scalable infrastructure and resource planning
- **SQA Role:** Track response times, throughput, and system stability; validate performance benchmarks (e.g., dashboard load time, transaction latency)

7. Security Assurance & Regulatory Compliance:

Critical for maintaining **user trust and legal compliance** in a financial application.

- **Development Team:** Implement secure authentication (OTP, biometrics), encryption, and secure APIs
- **Test Team:** Conduct penetration testing, vulnerability scanning, and security validation
- **Managers:** Ensure compliance with financial regulations (KYC, data protection laws)
- **SQA Role:** Perform security audits, track vulnerabilities, ensure compliance documentation, and validate adherence to regulatory standards

8. Release Management & Quality Gates:

Ensures that only **stable, secure, and high-quality builds** are released to production.

- **Development Team:** Prepare release builds, documentation, and rollback strategies
- **Test Team:** Execute final regression testing and UAT validation

- **Managers:** Approve release based on readiness metrics
- **SQA Role:** Enforce strict quality gates (zero critical defects, acceptable performance levels, security clearance), conduct release audits, and ensure production readiness

9. Production Monitoring & Continuous Improvement

Focuses on **post-release quality tracking and continuous enhancement** based on real user behavior and feedback.

- **Development Team:** Fix production issues and implement improvements
- **Test Team:** Validate fixes and perform regression testing
- **Managers:** Prioritize enhancements and manage release cycles
- **SQA Role:** Monitor production metrics (crash rates, latency, errors), analyze user feedback, track SLA compliance, and drive process improvements

Final Insight

These process activities ensure that INDmoney consistently meets its **quality benchmarks in terms of accuracy, security, performance, and usability**. As an SQA, continuous monitoring, auditing, and optimization of these processes is essential to deliver a **trustworthy, scalable, and high-quality financial platform**, where even minor defects can have significant business and user impact.

Release Criteria for INDmoney (5th deliverable)

Release criteria define the **minimum quality conditions that must be satisfied before deploying the application to production**. For a finance application like INDmoney, release decisions must be **strict, data-driven, and risk-aware**, as any failure can directly impact user finances, trust, and regulatory compliance. As an SQA, these criteria act as **quality gates** to ensure only stable, secure, and reliable builds are released.

1. Functional Completeness & Requirement Coverage:

All planned features and user stories must be **fully implemented and validated** against requirements.

- **Criteria:** 100% completion of committed features for the release
- **Validation:** All acceptance criteria passed; Requirement Traceability Matrix (RTM) shows full coverage
- **SQA Check:** No missing or partially implemented critical functionality

2. Zero Critical & High Severity Defects:

The application must be free from **critical and high-impact defects**, especially those affecting financial transactions or data accuracy.

- **Criteria:**
 - 0 Critical defects
 - 0 High severity defects
 - Medium/Low defects within acceptable limits
- **SQA Check:** Defect triage reports validated; no known production blockers

3. Data Accuracy & Financial Integrity Validation:

All financial data (portfolio values, transactions, balances) must be **accurate and consistent across systems**.

- **Criteria:** $\geq 99.99\%$ data accuracy verified
- **Validation:** Data reconciliation across APIs, databases, and UI
- **SQA Check:** No mismatch in financial calculations or reporting

4. Performance & Scalability Benchmarks Met:

The application must meet defined **performance standards under expected and peak loads**.

- **Criteria:**
 - Response time < 2 seconds for key operations
 - Stable performance under peak load conditions
- **Validation:** Load, stress, and endurance testing passed
- **SQA Check:** No performance degradation in critical workflows

5. Security Clearance & Vulnerability-Free Build:

The build must pass all **security validations**, ensuring no risk to user data or transactions.

- **Criteria:**
 - Zero critical/high security vulnerabilities
 - Secure authentication and encryption validated
- **Validation:** Penetration testing, vulnerability scans
- **SQA Check:** Security audit sign-off completed

6. Compliance & Regulatory Approval:

The release must comply with **financial regulations, KYC norms, and data privacy**

- **Criteria:** 100% compliance with applicable standards
- **Validation:** Audit reports and compliance checklists verified
- **SQA Check:** Legal and regulatory approvals obtained

7. Successful Regression & UAT Sign-off:

All existing and new features must be **validated through regression testing and User Acceptance Testing (UAT)**.

- **Criteria:**
 - 100% pass rate for critical test cases
 - Business stakeholders approve UAT
- **Validation:** Regression suite execution, UAT reports
- **SQA Check:** Formal sign-off from stakeholders

8. Production Readiness & Deployment Validation:

The system must be **fully prepared for deployment**, including infrastructure and rollback mechanisms.

- **Criteria:**
 - Deployment scripts validated
 - Rollback plan tested
 - Monitoring and alerting enabled
- **Validation:** Pre-production environment testing
- **SQA Check:** Release readiness checklist completed

9. Post-Release Monitoring Plan in Place:

A mechanism must be established to **monitor system health and user impact after release**.

- **Criteria:**
 - Real-time monitoring enabled (errors, crashes, performance)
 - Support team readiness confirmed
- **Validation:** Logging, alerting systems tested
- **SQA Check:** Incident response plan in place

Final Insight

The release criteria for INDmoney ensure that every deployment is **secure, stable, compliant, and reliable**. As an SQA, enforcing these strict quality gates is essential to protect **user trust, financial accuracy, and business reputation**.

Team Members and Division of Work:

Team members	Division of work
Salandri Shirisha :2025TM93131	Introduction,Generic SQA function and brainstorming discussion inputs (1st deliverable)
Sakthi P: 2025TM93220	Generic SQA functions and doc editing (1st deliverable)
Sanjana Nayak: 2025TM93056	Overview of project,Specific features (2nd deliverable)
Sajja Chandana: 2025TM93117	Added Specific features ,and meeting conduct (2rd deliverable)
Saloni Kashyap: 2025TM93083	Quality Requirements & inputs (3th deliverable)
Sanika Vineet kshirsagar: 2025TM93040	Process Activities & group creation (4th deliverable)
Sanjai Siddharthan M: 2025TM93014	Process Activities,flow design (4st deliverable)
Saif Uddin Ahamed: 2025TM93168	Release Criteria & final edits (5st deliverable)

Thank you. :)